BUSINESS REPORT

Olist E-Commerce Data Analysis: Customer Satisfaction and Lead Conversion

Project: Olist E-Commerce Capstone Project
Tools Used: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

1. **Executive Summary**

This report analyzes customer satisfaction, product-category reviews, lead conversion performance, and delivery times using Olist e-commerce data. The objective is to identify patterns that can help the business improve customer experience, optimize marketing performance, and make informed operational decisions.

The analysis covered 99,224 customer reviews and found an average review score of 4.09 out of 5. Five-star reviews accounted for approximately 57.8% of all reviews, indicating generally positive customer feedback. However, approximately 14.7% of reviews received one or two stars, suggesting opportunities to improve customer satisfaction.

The lead conversion analysis showed that paid search had the highest conversion rate among identifiable sources at 12.30%, followed by organic search at 11.80% and direct traffic at 11.22%. Organic search generated the largest number of closed leads, with 271.

The product-category analysis identified categories with high numbers and proportions of low-rated reviews. The delivery-time analysis showed that average review scores decreased as delivery time increased, falling from 4.41 for orders delivered within seven days to 2.18 for orders taking 31 days or more.

The report recommends improving delivery performance, investigating negative customer feedback, evaluating marketing channels using both conversion rates and lead volumes, and monitoring customer satisfaction consistently.

2. **Introduction and Business Problem**

Customer satisfaction and effective marketing are essential to the success of e-commerce businesses. Positive customer experience can encourage repeat purchases and strengthen customer trust, while effective marketing channels help businesses convert qualified leads into customers.

However, businesses may experience challenges such as delivery delays, product-related complaints, and marketing channels with low conversion rates. Understanding these issues requires analyzing customer reviews, order information, product categories, and lead conversion data.

The main business problem addressed in this project is: What factors influence customer satisfaction and lead conversion success?

The project examines four questions:

1. What is the distribution of customer review scores?
2. Which lead sources have the highest conversion rates?
3. Which product categories receive the most negative reviews?
4. Are longer delivery times associated with lower review scores?

The findings are intended to support decisions that improve customer experience, operational performance, and marketing effectiveness.

3. Data and Methodology

The analysis was conducted using Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook.

The CSV datasets were loaded into Pandas Data Frames and inspected to understand their structures, columns, data types, and missing values. Data preparation included checking duplicates, examining missing information, and preparing the datasets for analysis.

Missing written review comments were not automatically treated as invalid records because customers can provide numerical ratings without writing comments.

For customer satisfaction analysis, review scores were counted and summarized, and the average review score was calculated.

For lead conversion, the conversion rate was calculated using the formula:

Conversion Rate (%) = (Closed Leads / Qualified Leads) × 100

Lead sources were grouped and ranked by conversion rate. The unknown source was excluded from the comparison of identifiable marketing channels because its origin could not be determined.

For product-category analysis, review information was linked with order-item and product data. Reviews with scores of one or two were classified as low-rated reviews. Both the number of low-rated reviews and the proportion of reviews receiving low scores were examined. A minimum of 100 reviews was used for the low-review-rate comparison.

For delivery analysis, delivery time was calculated using purchase and actual delivery timestamps. Orders were grouped into five delivery-time categories, and the number of

orders and average review score were calculated for each group.

Charts were created to communicate the results and make patterns easier to interpret.

4. **Results and Findings**

4.1 **Customer Review Score Distribution**

The review dataset contained 99,224 reviews, with an average score of approximately 4.09 out of 5.

Review score	Number of reviews	Percentage
1 star	11,424	11.51%
2 stars	3,151	3.18%
3 stars	8,179	8.24%
4 stars	19,142	19.29%
5 stars	57,328	57.78%

Five-star reviews were the most common, representing approximately 57.8% of all reviews. One- and two-star reviews combined accounted for approximately 14.7%.

These findings indicate that customer feedback was generally positive. Nevertheless, the number of low-rated reviews suggests that some customers experienced dissatisfaction. Investigating these reviews may help identify recurring problems with products, sellers, delivery, or customer expectations.

4.2 **Lead Conversion Analysis**

The lead conversion analysis produced the following ranking:

Rank	Lead source	Qualified leads	Closed leads	Conversion rate
1	Paid search	1,586	195	12.30%
2	Organic search	2,296	271	11.80%
3	Direct traffic	499	56	11.22%
4	Referral	284	24	8.45%
5	Social	1,350	75	5.56%
6	Display	118	6	5.08%
7	Other publicities	65	3	4.62%
8	Email	493	15	3.04%
9	Other	150	4	2.67%

Paid search achieved the highest conversion rate among the identifiable sources at 12.30%. Organic search followed at 11.80% and generated the largest number of closed leads, with 271. Direct traffic also performed strongly at 11.22%.

Email and other sources had the lowest conversion rates, at 3.04% and 2.67%, respectively. The unknown source recorded a conversion rate of 16.65% in the initial analysis but was excluded from the ranking of identifiable channels.

These results suggest that paid search, organic search, and direct traffic are promising sources for acquiring customers. However, conversion rate alone does not measure profitability. Marketing costs, revenue, and customer acquisition costs should also be considered before reallocating budgets.

4.3 **Product Categories and Negative Reviews**

The product-category analysis examined reviews with scores of one or two and connected them with product categories.

The chart of low-rated review counts highlighted categories including bed, table and bath; health and beauty; furniture and decor; computers and accessories; sports and leisure; watches and gifts; household utilities; telephony; automotive; and toys.

Bed, table and bath had the largest number of low-rated review associations among the categories displayed.

The low-review-rate analysis highlighted categories such as men's fashion, office furniture, audio, construction tools and safety, and home comfort among categories with relatively high proportions of low-rated reviews, subject to the minimum review-count filter.

These two measures provide different insights. A category may have many negative reviews because it receives many orders, while another category may have a higher proportion of dissatisfied customers despite receiving fewer reviews.

The business should investigate both measures and examine customer comments, product descriptions, seller performance, and delivery issues to understand the causes of dissatisfaction.

4.4 **Delivery Time and Customer Satisfaction**

The delivery-time analysis produced the following results:

Delivery time| Number of orders| Average review score
0–7 days| 33,541| 4.41
8–14 days| 36,191| 4.29
15–21 days| 15,264| 4.10
22–30 days| 6,826| 3.49
31+ days| 4,008| 2.18

The average review score declined as delivery time increased. Orders delivered within seven days received an average score of 4.41, while orders taking 31 days or more averaged 2.18.

The decline became more pronounced after 21 days. The difference between the fastest and slowest delivery groups was 2.23 points on the five-point review scale.

This is a strong negative association between delivery time and customer satisfaction. It suggests that long delivery delays may contribute to poor customer experiences. However, the analysis does not prove that delivery time alone causes lower scores, because product quality, seller performance, and customer expectations may also influence ratings.

5. **Business Recommendations**

Based on the findings, the following recommendations are proposed.

5.1 Improve Delivery Performance

The business should investigate orders with extended delivery times, identify recurring logistics bottlenecks, and improve delivery tracking and customer communication. Customers should be notified proactively when orders are delayed.

Delivery time, delayed-order frequency, and average review scores should be monitored together to evaluate improvement.

5.2 **Investigate Negative Reviews**

Product categories with high numbers or proportions of low-rated reviews should receive further investigation. The business should examine written complaints, identify recurring product or seller issues, and check whether product descriptions accurately represent the products sold.

Both negative-review counts and negative-review rates should be considered when prioritizing categories.

5.3 Optimize Marketing Channels

Paid search, organic search, and direct traffic should be evaluated closely because they have relatively high conversion rates. The business should investigate which campaigns and activities generate qualified leads and closed customers.

Email and other lower-converting channels should be reviewed to determine whether targeting, messaging, or follow-up processes can be improved. Marketing budgets should be allocated using conversion performance, lead volume, acquisition cost, and revenue rather than conversion rate alone.

5.4 **Monitor Customer Satisfaction**

The business should regularly track average review scores, the proportion of one- and two-star reviews, delivery performance, and negative-review patterns by category and seller.

Consistent monitoring can help management identify emerging problems and evaluate whether corrective actions improve the customer experience.

6**. Limitations**

The findings identify patterns and associations rather than proving causation. Factors such as product quality, seller performance, customer expectations, and order characteristics may also influence review scores.

Negative-review counts depend partly on the number of orders and reviews received by each category. Therefore, counts should be interpreted alongside low-review rates.

Lead conversion rates do not measure marketing profitability, as the analysis did not include campaign costs, revenue, or customer lifetime value.

The delivery analysis also depends on accurate timestamps, appropriate data cleaning, and correct dataset merging. Order counts should be verified to ensure that duplicate rows have not inflated the results.

7. **Conclusion**

This project used data analysis and visualization to examine customer satisfaction, product-category review patterns, lead conversion performance, and delivery time.

The review analysis found an average score of 4.09 out of 5, with five-star reviews representing approximately 57.8% of all reviews. Paid search achieved the highest conversion rate among identifiable lead sources at 12.30%, while organic search generated the largest number of closed leads.

The product-category analysis highlighted areas for further investigation into negative customer feedback. The delivery-time analysis showed that average review scores decreased from 4.41 for orders delivered within seven days to 2.18 for orders taking 31 days or more.

Overall, the results suggest that improving delivery performance, investigating recurring product-related complaints, and evaluating marketing channels using conversion rates alongside costs and lead volumes could help improve business performance.

The project demonstrates how data cleaning, analysis, and visualization can transform raw data into practical insights that support better business decisions.